

Economic Forecasting for Banking Using FRED Time Series: Gradient Boosting with Phased Risk Scenario Analysis

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GitHub repository: https://github.com/lynchpo/dragon_exploder_fred

Background & Question

Banks and financial institutions rely on accurate forecasts of key interest rates to manage lending risk, set product pricing, and conduct stress testing. Among these rates, the 30-year fixed mortgage rate is one of the most consequential benchmarks in the U.S. housing market. Movements in this rate directly affect borrowing costs for households, refinancing volumes, and the valuation of mortgage-backed assets held by lenders. Even modest improvements in the ability to anticipate rate changes can therefore produce meaningful operational and financial benefits for risk-management and treasury teams.

Research question: Can a carefully engineered set of macroeconomic time series from the Federal Reserve Economic Data (FRED) repository be used to produce accurate multi-step forecasts of the 30-year fixed mortgage rate while also generating scenario-based risk metrics that support decision-making under uncertainty?

This question is interesting and important for three reasons. First, mortgage rates remain a central transmission channel of monetary policy into the real economy; improved forecasts therefore have both private and public value. Second, the problem sits at the intersection of classical time-series econometrics and modern machine-learning methods, offering a clear opportunity to compare approaches under realistic constraints. Third, the availability of high-frequency, publicly maintained data through FRED makes the problem reproducible and suitable for an end-to-end data-science workflow. The primary stakeholders are risk-management, treasury, and mortgage-product teams at commercial banks and specialized mortgage lenders who must continuously update rate assumptions for pricing, capital planning, and stress tests.

Prior research has mapped both the economic drivers of mortgage rates and the growing usefulness of machine-learning methods for interest-rate forecasting. Early econometric studies established that Treasury yields, inflation, and labor-market conditions exert a strong influence on mortgage rates. More recent work has shown that gradient-boosting models frequently exceed traditional statistical approaches on interest-rate series (Chen et al. 2024; Kim and Park 2024). The Federal Reserve Bank of Dallas has specifically tested machine-learning techniques for nowcasting and forecasting U.S. mortgage rates with FRED data (Federal Reserve Bank of Dallas 2023). Complementary research has examined Monte Carlo simulation as a way to move beyond point forecasts and improve risk assessment (Liu et al. 2025). The present project builds on these contributions by constructing a clean, reproducible pipeline that (1) systematically expands and selects features from the large FRED universe, (2) compares classical and modern forecasting models under proper time-series validation, and (3) treats full Monte Carlo scenario generation as a phased extension that can be completed after the core forecasting pipeline is validated.

Hypothesis: A carefully selected combination of leading macroeconomic indicators contains predictive information about future movements in the 30-year fixed mortgage rate because these indicators capture the primary channels through which monetary policy and real-economy conditions influence long-term borrowing costs.

Prediction: A gradient-boosting model that incorporates engineered lag, rolling, and yield-curve features will achieve lower out-of-sample forecasting error (RMSE and MAE) than a SARIMAX baseline. When Monte Carlo paths are generated from the fitted model (Phase 2), the resulting distribution of outcomes will support calculation of risk metrics such as maximum drawdown that add decision-relevant information beyond the point forecast alone.

Data & Methods

Data will be downloaded from the Federal Reserve Economic Data (FRED) repository at the Federal Reserve Bank of St. Louis (<https://fred.stlouisfed.org/>). FRED is the standard public source for U.S. macroeconomic time series and is commonly used in both academic research and industry forecasting. Series are obtained programmatically

through the fredapi Python package, which returns pandas DataFrames and supports reproducible acquisition scripts. Because FRED series are revised over time, each pull will record the observation date and the vintage of the data; this information will be retained for auditability.

The target variable is the 30-Year Fixed Rate Mortgage Average (FRED series MORTGAGE30US), a weekly series that reflects the average contract rate on 30-year fixed-rate mortgages. An initial core set of five predictor series has been selected on the basis of established economic relationships and prior forecasting literature: the 10-Year Treasury Constant Maturity Rate (DGS10), the Unemployment Rate (UNRATE), the Consumer Price Index for All Urban Consumers (CPIAUCSL), Housing Starts (HOUST), and the Federal Funds Effective Rate (FEDFUNDS). These five series capture the principal channels of monetary policy transmission, inflation, labor-market slack, and housing-market activity. Because FRED contains hundreds of potentially relevant indicators, the project explicitly plans a systematic expansion of the feature space. Additional candidate series (for example, measures of credit spreads, consumer sentiment, money-supply aggregates, and alternative inflation gauges) will be screened through correlation analysis, lagged cross-correlation, and simple univariate predictive power. The final feature set will be determined by a combination of domain knowledge, statistical screening, and model-based importance ranking rather than by an a-priori limit of five series. This staged approach addresses the concern that an arbitrarily small feature space may omit important information while still keeping the initial modeling pipeline tractable.

A central technical challenge is the presence of mixed sampling frequencies: MORTGAGE30US is weekly, DGS10 is daily, and UNRATE and CPIAUCSL are monthly. All series will be aligned to a common weekly frequency using pandas resample operations followed by forward-filling (ffill). Forward-filling is applied only to lagging series and is performed strictly within the information set available at each forecast origin, thereby avoiding look-ahead bias. After alignment, lag features (1-, 2-, and 4-week lags of both the target and selected predictors), rolling statistics (4- and 12-week means and standard deviations), and yield-curve spreads (for example,

DGS10 minus FEDFUNDS) will be constructed. These engineered features supply the momentum and relative-price signals that tree-based models require.

The historical window will extend from approximately 2000 through the most recent available observations, deliberately spanning multiple interest-rate regimes (pre-crisis, post-crisis low-rate environment, and the 2022–2023 tightening cycle). This breadth is essential because tree-based models such as XGBoost and LightGBM cannot generate predictions outside the range of values present in the training data; they simply do not extrapolate. Training on a wide historical distribution that includes both high-rate and low-rate periods therefore reduces the risk that future rate levels will fall outside the model's observed support. All preprocessing, feature construction, and model fitting will be implemented inside scikit-learn Pipelines and time-series cross-validation loops provided by the skforecast library so that the entire workflow remains reproducible and free of leakage.

Analysis Plan

The analysis is organized in two phases to ensure that a complete, evaluated forecasting pipeline is delivered within the Capstone window while still leaving room for more advanced risk analytics.

Phase 1: **Core Forecasting Pipeline (Capstone deliverable).**

1. Additional preprocessing: complete frequency alignment, missing-value diagnostics, outlier checks, and construction of the full lag/rolling/spread feature set described above.
2. Exploratory analysis: time-series plots of the target and core predictors, correlation heat-maps, lagged cross-correlation functions, and simple univariate predictive screens to guide feature expansion.
3. Baseline model: SARIMAX (statsmodels) fitted on the aligned weekly series.
4. Additional models: XGBoost and LightGBM regressors with time-series cross-validation (skforecast). Hyper-parameters will be tuned via grid or randomized search inside the time-series CV loop.
5. Evaluation: RMSE, MAE, and directional accuracy on walk-forward and expanding-window validation sets. SHAP values will be computed to assess feature importance and to test whether the hypothesized macroeconomic channels appear among the top predictors.
6. Success criterion for the research question: a gradient-boosting model that materially improves upon the SARIMAX baseline under proper time-series

validation, together with interpretable evidence that leading indicators contribute predictive power.

Phase 2: Scenario Generation and Risk Metrics (stretch / post-Capstone if needed).

Once the core forecasting pipeline is validated, Monte Carlo paths will be generated from the preferred model to produce distributions of future mortgage-rate trajectories.

From these paths, maximum drawdown and related risk metrics will be calculated.

Presenting the work as phased ensures that a complete, evaluated forecasting system is delivered by the Capstone deadline even if the more computationally intensive simulation component requires additional time after the course ends.

Technical stack: Python; pandas and fredapi for data acquisition; scikit-learn Pipelines for preprocessing; statsmodels for SARIMAX; xgboost / lightgbm for gradient boosting; skforecast for time-series cross-validation and recursive forecasting; shap for interpretability; matplotlib / seaborn for visualization. Development will be conducted in Google Colab with all code version-controlled in the GitHub repository listed above.

The phased design, the explicit plan for feature-space expansion, and the use of a mature open-source stack together demonstrate that the proposed work is realistic within an approximately six-week Independent Study window.

Data Dictionary

The table below lists the core series that form the initial feature set. Additional FRED series will be evaluated and may be added after the systematic screening described in the Data & Methods section.

Variable / Series	Description	Frequency / Type	Role in Analysis
MORTGAGE30US	30-Year Fixed Rate Mortgage Average	Weekly / float	Target (response) variable
DGS10	10-Year Treasury Constant Maturity Rate	Daily / float	Predictor (yield-curve level)
FEDFUNDS	Federal Funds Effective Rate	Monthly / float	Predictor (policy rate)
UNRATE	Civilian Unemployment Rate	Monthly / float	Predictor (labor-market slack)
CPIAUCSL	Consumer Price Index for All Urban Consumers	Monthly / float	Predictor (inflation)
HOUST	Housing Starts: Total New Privately Owned	Monthly / float	Predictor (housing activity)
Yield-curve spread	DGS10 – FEDFUNDS (constructed)	Weekly / float	Engineered predictor (term premium / slope)

Lag features	1-, 2-, 4-week lags of target and selected predictors	Weekly / float	Engineered predictors (momentum)
Rolling statistics	4- and 12-week means and standard deviations	Weekly / float	Engineered predictors (local trend / volatility)

Works Cited

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